Bulk-Informed Cross-Modality Representation Learning for scRNA-seq Batch Correction

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Abstract

Single-cell RNA-seq (scRNA-seq) batch correction is the dominant pre-processing step in every modern single-cell pipeline, but a recent 2025 calibration audit found that seven of eight widely used methods introduce artefacts even under null conditions, and that evaluation metrics themselves are unreliable. Meanwhile, bulk RNA-seq atlases (GTEx, TCGA, ENCODE) contain orders of magnitude more donor-level coverage, cleaner technical noise, and decades of curated biological priors—and no current integration method uses them. We propose a cross-modality variational autoencoder with an adversarial modality discriminator that learns a shared, technology-invariant latent space for bulk and scRNA-seq, regularised by a *gene-discordance* supervision signal derived from matched bulk / pseudobulk pairs. The learned representation is evaluated primarily on scIB-style integration benchmarks (kBET, LISI, ASW, graph connectivity) and on null-condition calibration tests, with cell-type annotation transfer and bulk deconvolution as secondary downstream probes. The full pipeline trains in under two hours on a single consumer GPU.

1 Motivation

Single-cell RNA-seq resolves tissue into individual transcriptomes but is sparse ($\sim$80% zeros), shallow, and subject to large technology-specific biases: platform, chemistry version, sequencing depth, dissociation stress, and ambient RNA. Across 15 studies PC1 correlates with detection rate (r=0.54–0.93), cells cluster by lab rather than by type, and the first Alzheimer's snRNA-seq study reported 1,031 DEGs that collapsed to $\sim$16 after proper pseudobulk reanalysis (549-fold false-discovery inflation). Batch correction is therefore the critical bottleneck for every downstream scRNA-seq analysis—yet a 2025 null-condition audit shows 7 of 8 widely deployed methods create artefacts when no batch effect exists; only Harmony passes, and silhouette-based metrics are themselves unreliable. Foundation models such as Wen et al. (2024) and Levine et al. (2024) are trained purely on scRNA-seq and inherit its technical biases; neither incorporates bulk at any stage.

Bulk RNA-seq is the obvious untapped prior. GTEx v11 alone provides $\sim$17,000 deeply sequenced bulk samples across 54 tissues, with cleaner noise and no dissociation artefacts. Matched bulk/pseudobulk comparisons expose $\sim$557 reproducibly discordant genes (Hu, 2026) whose distributions disagree even after identical normalisation—a measurable, gene-resolved signature of sc technical bias. No published integration method uses bulk-derived gene-level variance, expression distributions, or discordance signals as priors for scRNA-seq batch correction. This proposal closes that gap.

Contributions. (i) A dual-encoder variational architecture with an adversarial modality classifier that aligns bulk and scRNA-seq log-CPM distributions in a shared d=64 latent space, producing batch-corrected embeddings anchored by the cleaner bulk modality. (ii) A *gene-discordance* auxiliary head that learns a per-gene bulk–sc disagreement score, supervised on matched donor pairs, which acts as an explicit inductive bias against integrating sc batches along bias directions bulk contradicts. (iii) A rigorous calibration-aware evaluation protocol: scIB integration metrics, the 2025 null-condition random pseudo-batch test, and downstream probes (label transfer, pseudobulk DE reproducibility vs. matched bulk, deconvolution) that we expect to improve together if the representation is honest, and disagree if any metric is gameable. (iv) Systematic ablations (no adversarial loss, no bulk modality, no discordance head) that quantify which ingredient drives each metric.

2 Problem Definition

Let $X \in \mathbb{R}^{G\times N}$ be an scRNA-seq matrix over G shared genes and N cells, with per-cell batch labels $m \in \{1, \ldots, M\}^N$ and optional cell-type labels $y \in \{1, \ldots, K\}^N$. Let $B \in \mathbb{R}^{G\times D}$ be a bulk matrix from D donors (possibly different donors from X). We want to learn an encoder $E : \mathbb{R}^G \to \mathbb{R}^d$ such that, for a cell x_i, $z_i = E(x_i)$ satisfies three desiderata simultaneously:

1. **Batch-invariant:** z is statistically independent of m conditional on biology; formally, a classifier trained to predict m from z achieves AUC near chance on a null pseudo-batch split.
2. **Biology-preserving:** cells of the same type cluster together across batches; $\mathrm{ASW}(z, y)$ is high on held-out batches.
3. **Bulk-aligned:** bulk profiles embed into the *same* latent space and nearby regions correspond to biologically matched tissues, giving bulk and sc a common coordinate system.

This last requirement is what makes the problem bulk-*informed* rather than merely another sc integration model: it imports bulk's cleaner technical noise as a prior and makes cross-modality transfer (sc $\leftrightarrow$ bulk) a built-in capability rather than a downstream bolt-on.

3 Method

Three coupled modules, trained end-to-end.

(1) Cross-modality VAE. Dual encoders $E_{\text{bulk}}, E_{\text{sc}} : \mathbb{R}^G \to \mathbb{R}^{2d}$ (MLPs, hidden $1024 \to 512$) output $(\mu, \log\sigma^2)$; reparameterise to $z \in \mathbb{R}^d$ with d=64. A shared decoder $D : \mathbb{R}^d \to \mathbb{R}^G$ reconstructs the input. A *modality discriminator* $C_{\text{mod}} : \mathbb{R}^d \to \{0, 1\}$ (2-layer MLP) classifies z as bulk or sc; a *batch discriminator* C_{batch} classifies z among the M sc batches. The encoders are trained adversarially against both discriminators. The encoder loss is

$$\mathcal{L}_{\text{enc}} = \mathcal{L}_{\text{recon}} + \beta(t)\,\mathcal{L}_{\text{KL}} - \lambda_m(t)\,\mathcal{L}_{\text{mod}} - \lambda_b(t)\,\mathcal{L}_{\text{batch}},$$

with $\beta, \lambda_m, \lambda_b$ annealed from 0 to avoid posterior collapse and early discriminator domination. When cell-type labels are available we optionally add a scANVI-style classifier on z as a semi-supervised anchor (an ablation; the default model is unsupervised).

(2) Gene-discordance head. A small MLP $H_d : \mathbb{R}^d \to \mathbb{R}^G$ predicts, from an sc cell's latent code z, a per-gene *discordance score* $\hat{d}_g \in \mathbb{R}$ that estimates how much that gene's value in the sc modality disagrees with the bulk estimate of the same biology. Supervision comes from matched donors: for each donor with paired bulk and pseudobulk, we compute $d_g = |\log \mathrm{FC}(b_g, \mathrm{pseudobulk}(X_g))|$ and regress $\hat{d}_g$ against d_g. This gives the encoder a gradient that explicitly distinguishes *genes whose sc values can be trusted* from *genes whose sc values systematically deviate from bulk*, and pushes the latent space to downweight the latter. The head is auxiliary—it is not required at inference.

(3) Bulk-prior regulariser. We regularise the latent distribution of sc cells toward the bulk latent distribution at the donor / tissue level, using an MMD term $\mathrm{MMD}^2\left(\{z_i^{\text{sc}}\}_{y_i=t}, \{z_j^{\text{bulk}}\}_{\text{tissue}(j)=t}\right)$ summed over tissues / cell types t. This makes bulk's cleaner gene-coexpression structure act as a distributional prior on the sc cells, without requiring per-cell ground truth.

Training schedule. *(a) Pretraining:* encoders, decoder, and both discriminators on unpaired bulk (GTEx, $\sim$800 donors) and sc (HCA, $\sim$390k cells) with only the VAE + adversarial losses, to shape an aligned latent space. *(b) Fine-tuning:* discordance head and MMD regulariser added, trained jointly on paired matched donors. No pretrained foundation models are used; all parameters ($\sim$6M total) are trained from scratch.

Why each component is needed. A naive VAE on the union of bulk and sc cleanly separates the two into disjoint latent clusters (the reconstruction objective alone gives no alignment pressure). The discordance head supplies an explicit gradient distinguishing biological batch differences from technology-specific ones (the 2025 audit's central failure mode). The MMD term imports the bulk prior at the *distributional* level, which the binary discriminator cannot. The three components target three distinct failure modes.

4 Datasets and Evaluation Protocol

4.1 Datasets

Pretraining (unpaired). *Bulk:* GTEx v11 whole-blood subset, $\sim$803 donors, filtered to $\sim$16k genes expressed in $\geq$10% of donors at CPM $\geq$ 1. *Single-cell:* HCA blood scRNA-seq, $\sim$390k cells, downsampled to a balanced 50k-cell pretraining set stratified across donors and cell types.

Batch-correction benchmark (primary evaluation). A scIB-style multi-batch cohort constructed from HCA blood and published 10x/Smart-seq2 pairs. Each cell has batch, donor, and cell-type labels, so both batch-mixing and biology-preservation can be scored.

Held-out matched bulk/sc cohort (discordance ground truth). The BAL paired benchmark of Hu (2026) (30 donors with matched bulk and sc) supplies the 557 published discordant genes used as validation for the gene-discordance head.

Downstream probe datasets. *(i) Cell-type transfer:* pancreatic islet scRNA-seq (GSE84133, Baron et al.) across 14 donors, label-transferred from a held-out donor. *(ii) Pseudobulk DE vs. real bulk:* sc $\to$ pseudobulk DE on a held-out cohort, compared to matched real bulk DE via rank concordance. *(iii) Bulk deconvolution:* GSE50244 pancreas bulk (89 donors with HbA1c) deconvolved with an NNLS solver applied on top of signatures derived from the corrected sc embeddings, reported as a sanity check that the latent space remains quantitatively linked to expression magnitudes.

4.2 Evaluation Protocol

Splits. Donors and batches are split 70/15/15 train/val/test. All cells from a test donor or a test batch are *never* seen during training (donor- *and* batch-level leakage control).

Primary metrics (mean $\pm$ std, 5 seeds). Batch mixing: kBET, iLISI, graph connectivity, batch ASW. Biology conservation: cLISI, cell-type ASW, Leiden NMI/ARI, isolated-label F1. We also report the scIB composite (since it alone masks overcorrection) and the 2025 null-condition calibration test (fraction of spurious clusters under a random pseudo-batch split—an honest integrator creates none).

Secondary / downstream metrics. Cell-type label transfer accuracy and macro-F1 from held-out donors; pseudobulk-vs-real-bulk DE rank concordance (Spearman on $|\log \mathrm{FC}|$) and discordant-gene recall on the BAL cohort; deconvolution RMSE/MAE/CCC on GSE50244 as a quantitative-fidelity probe (not the primary metric).

Baselines (fair comparison). All baselines consume the *same* preprocessed expression matrix, share the same train/val/test splits, and are evaluated with identical metrics on identical test sets: Harmony (the 2025 audit's only well-calibrated method), Seurat v5 RPCA, scVI, scANVI, Scanorama, and ComBat-seq. Ablations isolate each contribution: (a) no adversarial modality loss, (b) no bulk modality (sc-only encoder), (c) no discordance head, (d) no MMD regulariser.

5 Estimated Computing Resources

Hardware. One consumer GPU (NVIDIA RTX 3070/4070 class, 8–12 GB VRAM) or a free Google Colab T4. **Memory.** The full HCA blood sub-sample (50k cells $\times$ 16k genes at fp32) is $\sim$3 GB. **Runtime.** (i) Data preprocessing: $<$10 min on CPU. (ii) Cross-modality VAE pretraining: $\sim$60 min (100 epochs, batch 256). (iii) Joint fine-tuning with discordance and MMD heads: $\sim$30 min. (iv) Full 5-seed sweep across all baselines, metrics, and ablations: $<$10 GPU-hours end-to-end. **Storage.** Raw datasets $\sim$10 GB; processed artefacts $\sim$1.5 GB.

The project uses *no* pretrained foundation models, so compute does not scale with any external checkpoint. All code is PyTorch, scikit-learn, scanpy, and scib, already installed and exercised in the existing project repository. The end-to-end pipeline is comfortably executable within the remaining course window with headroom for additional ablations.

Timeline. *Weeks 1–2:* assemble the scIB cohort, reproduce Harmony/scVI/scANVI baselines. *Weeks 3–4:* cross-modality VAE + adversarial trainer; verify null-condition calibration. *Weeks 5–6:* discordance head + MMD regulariser + ablations. *Week 7:* downstream probes, paired bootstrap tests, write-up. A one-week buffer remains before the deadline.

Risks and fallbacks. If adversarial training is unstable, swap the discriminator for pure MMD alignment. If the discordance head overfits the small matched cohort, freeze it as a fixed gene weighting. Both fallbacks preserve the core cross-modality integration contribution.

6 Related Work

Single-cell foundation models. Wen et al. (2024) pretrain a cell language model on scRNA-seq with strong imputation and perturbation-prediction performance. The model is scRNA-seq only: bulk

RNA-seq, with its cleaner noise and orders-of-magnitude larger donor coverage, is not used at any stage. The same is true of every foundation model in the 2025 survey. Our work is complementary: bulk is a first-class modality during pretraining, and the shared latent space is explicitly aligned by an adversarial loss rather than left to emerge.

LLMs for single-cell. Levine et al. (2024) tokenise scRNA-seq profiles as ranked gene sequences and apply an LLM. The gene-rank tokenisation discards absolute expression magnitudes that matter for batch correction (where detection rate itself is the dominant axis of variation). Our method keeps a continuous latent representation, so magnitude information and gradient-based integration losses remain available.

Cross-modal representation learning. Xu et al. (2023) survey alignment, fusion, and translation strategies for multimodal transformers, primarily in vision-language. We adopt their fusion-via-shared-latent paradigm but replace cross-attention alignment (appropriate for token-aligned modalities) with an adversarial modality classifier and an MMD distributional prior, because bulk and scRNA-seq have no natural token correspondence and differ primarily in their *distribution*, not in their feature layout.

Justified novelty. Relative to the three works above, our contribution is the first to combine the following in a single end-to-end batch-correction model: (i) bulk and scRNA-seq as first-class modalities in a shared-latent VAE, beyond the sc-only pretraining of CellPLM; (ii) an explicitly supervised gene-discordance head anchored on measurable bulk/pseudobulk disagreement, addressing the calibration failure mode ignored by both CellPLM and Cell2Sentence; and (iii) a distributional MMD prior that imports bulk's cleaner noise structure into the sc integration objective, a use of the shared-latent paradigm of Xu et al. (2023) that their vision-language setting does not need. Each ingredient has precedent in isolation, but the combination, targeted at the calibration crisis of scRNA-seq batch correction and anchored on the discordance signal that matched bulk/pseudobulk pairs make measurable, has not been reported in any of the allowed venues.

References

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